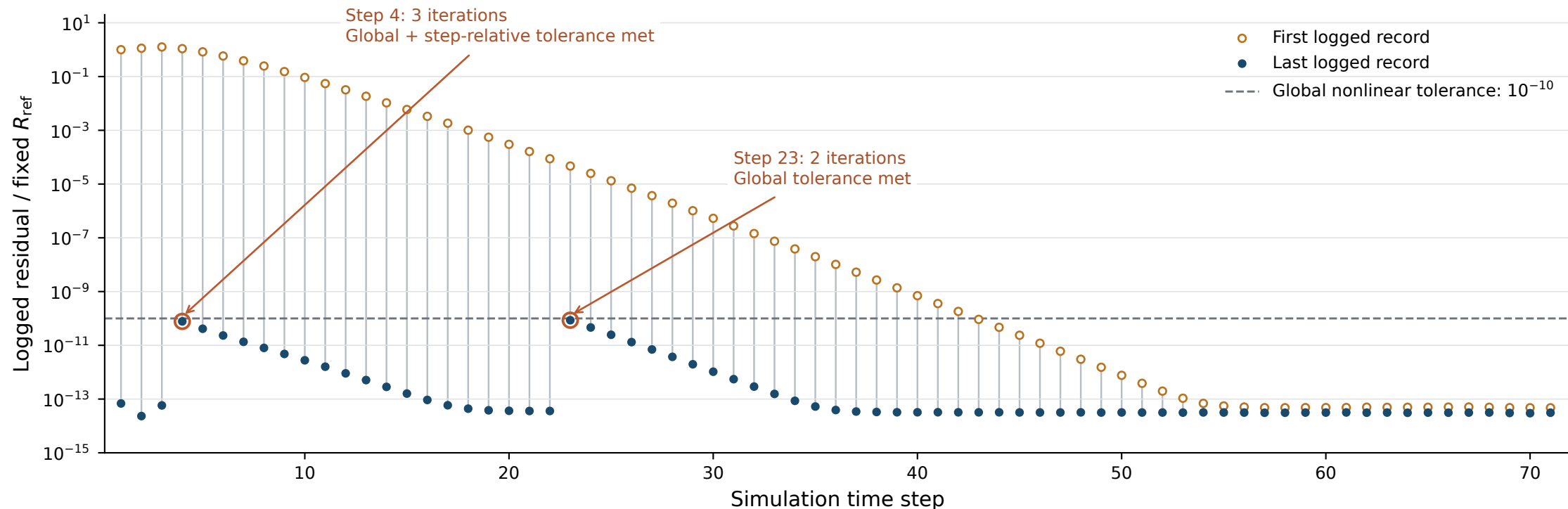


Nonlinear Residual Reduction Within Each Time Step

Each vertical connector pairs the first and last records of ONE step. $R_{\text{ref}} = R^{(1,1)}$ is fixed.



No lines connect last records across steps. At steps 4 and 23, fewer iterations explain the higher last-record values.

Neither event rebuilds the preconditioner. All steps satisfy the actual nonlinear stopping rule; the iteration cap is never reached.

The low residual plateau is not an established numerical floor or a physical-accuracy estimate.